

Friction • Worked Solutions

Odd-numbered problems 5.23-5.41 | Statements condensed; all parts included

Static friction adjusts up to $f_{s,\max} = \mu_s N$; sliding friction has magnitude $f_k = \mu_k N$. Constant velocity means net force is zero.

5.23 Maximum bed tilt

A patient lies on a bed with $\mu_s = 1.20$ between the patient and sheets. Find the largest angle to the floor before the patient starts sliding.

For the patient: weight mg acts downward, N is perpendicular to the bed, and static friction acts uphill. At the threshold of sliding:

$$N = mg \cos \theta; \quad mg \sin \theta = \mu_s N$$
$$\tan \theta = \mu_s = 1.20$$

Answer: $\theta_{\max} = \arctan(1.20) = 50.2^\circ$.

5.25 Pushing a box, then letting it coast

A 16.8 kg box moves at 3.50 m/s on a level surface with $\mu_k = 0.20$. (a) Find the horizontal push needed for constant speed. (b) Find its stopping distance after the push is removed.

(a) Vertically, $N = mg$. Horizontally, constant velocity requires the push to balance kinetic friction.

$$F = f_k = \mu_k mg = (0.20)(16.8)(9.80) = 32.928 \text{ N}$$

Answer (a): $F \approx 32.9 \text{ N}$ in the direction of motion.

(b) After release, only friction acts horizontally. Take the initial motion as positive.

$$a = -\mu_k g = -1.96 \text{ m/s}^2$$
$$v^2 = v_0^2 + 2a\Delta x$$
$$0 = (3.50)^2 - 2(1.96)\Delta x$$

Answer (b): $\Delta x = 3.125 \text{ m} \approx 3.13 \text{ m}$.

Force directions: N upward; mg downward; friction opposite motion; the applied push is present only in part (a).

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5.27 • Friction on Earth and the Moon | 5.29 • Playground slide

5.27 Crate of tools

A 45.0 kg crate starts moving when a horizontal push exceeds 313 N; a 208 N push then keeps it moving at 25.0 cm/s. (a) Find μ_s and μ_k . (b) Find the push for $a = 1.10 \text{ m/s}^2$. (c) On the Moon ($g = 1.62 \text{ m/s}^2$), find the starting push and the acceleration with the push from (b).

(a) $N = mg = (45.0)(9.80) = 441 \text{ N}$. Use the starting threshold for static friction and the steady push for kinetic friction.

$$\mu_s = 313/441 = \mathbf{0.710}; \quad \mu_k = 208/441 = \mathbf{0.472}$$

(b) Once sliding: $F - f_k = ma$, so $F = 208 + (45.0)(1.10)$.

Answer (b): $F = 257.5 \text{ N} \approx 258 \text{ N}$.

This is the force needed **while moving**. To start from rest on Earth, first exceed 313 N, then reduce the push to 257.5 N.

(c) Assume the friction coefficients are unchanged on the Moon. Use their unrounded values.

$$\begin{aligned} F_{\text{start,threshold}} &= \mu_s mg_{\text{Moon}} = 313(1.62/9.80) = \mathbf{51.7 \text{ N}} \\ f_{k,\text{Moon}} &= 208(1.62/9.80) = 34.38 \text{ N} \\ a &= (257.5 - 34.38)/45.0 = \mathbf{4.96 \text{ m/s}^2} \end{aligned}$$

A push just above 51.7 N starts it moving; the acceleration is in the direction of the push.

5.29 Slide: estimates and general formulas

Estimate a slide's height and angle. (a) Find the bottom speed from rest with negligible friction. (b) Without wax paper, the speed is half of (a); find μ_k . (c) A differently dressed child stays at rest; find the minimum μ_s .

Chosen estimates: vertical height $h = 2.0 \text{ m}$ and angle $\theta = 30^\circ$. These are assumptions, not values supplied by the problem. Ramp length $L = h/\sin \theta$.

(a) With negligible friction, $a = g \sin \theta$ and $v^2 = 2aL = 2gh$.

$$v = \sqrt{2gh} = \sqrt{2(9.80)(2.0)} = \mathbf{6.26 \text{ m/s}} \text{ (about } 6.3 \text{ m/s)}$$

(b) With friction, $v^2 = 2gL(\sin \theta - \mu_k \cos \theta)$. Half the original speed means one quarter of its squared speed:

$$\frac{1}{4} = 1 - \mu_k \cot \theta \Rightarrow \mu_k = \frac{3}{4} \tan \theta = \mathbf{0.433}$$

(c) Rest requires $mg \sin \theta \leq \mu_s mg \cos \theta$.

$$\mu_{s,\text{min}} = \tan \theta = \tan 30^\circ = \mathbf{0.577}.$$

Friction • Worked Solutions

5.31 • Forces along a ramp | 5.33 • Two boxes lowered together

5.31 Push down or up a ramp

A 10.0 kg box slides on a 55.0° ramp with $\mu_k = 0.300$. A 120.0 N force acts parallel to the ramp. Find the acceleration magnitude for (a) a downhill push and motion; (b) an uphill push and motion.

$N = mg \cos 55.0^\circ = 56.2 \text{ N}$, so $f_k = 16.86 \text{ N}$. The downhill weight component is $mg \sin 55.0^\circ = 80.28 \text{ N}$. Friction opposes the stated motion.

(a) Choose downhill positive:

$$ma = F + mg \sin \theta - f_k$$
$$a = (120.0 + 80.28 - 16.86)/10.0 = \mathbf{18.3 \text{ m/s}^2 \text{ downhill.}}$$

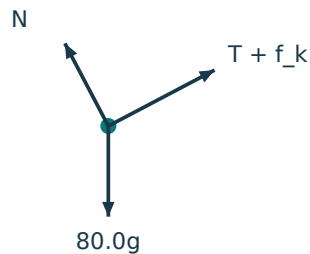
(b) Choose uphill positive:

$$ma = F - mg \sin \theta - f_k$$
$$a = (120.0 - 80.28 - 16.86)/10.0 = \mathbf{2.29 \text{ m/s}^2 \text{ uphill.}}$$

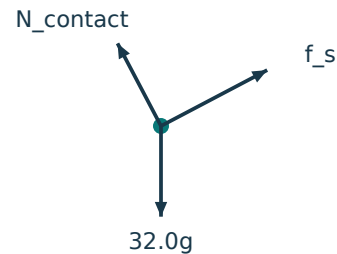
5.33 Stacked boxes moving down a ramp

A 32.0 kg box sits on a 48.0 kg box. Both descend together at 15.0 cm/s, held by a rope parallel to the ramp attached to the lower box. The ramp rises 2.50 m over a horizontal run of 4.75 m; $\mu_k = 0.444$ at the ramp and $\mu_s = 0.800$ between boxes. (a) Find the rope force. (b) Find the friction on the upper box.

Both boxes together



Upper box only



$\theta = \arctan(2.50/4.75) = 27.76^\circ$. Constant velocity means $a = 0$. The rope contributes one external tension to the two-box system; friction between boxes is internal.

(a) For both boxes together, $M = 80.0 \text{ kg}$. Choose uphill positive:

$$N = Mg \cos \theta; \quad T + \mu_k Mg \cos \theta - Mg \sin \theta = 0$$
$$T = Mg(\sin \theta - \mu_k \cos \theta) = \mathbf{57.1 \text{ N uphill}}$$

(b) For the upper box alone, static friction balances its downhill weight component:

$$f_s = (32.0)g \sin \theta = \mathbf{146 \text{ N uphill}}$$
$$f_{s,\text{max}} = (0.800)(32.0)g \cos \theta = 222 \text{ N}$$

Because $146 \text{ N} < 222 \text{ N}$, the upper box can move with the lower box without slipping. Static friction here is 146 N, not its maximum value.

Friction • Worked Solutions

5.35 • Stopping distance | 5.37 • Two crates pulled at constant velocity

5.35 Stopping with locked wheels

(a) With $\mu_k = 0.80$ on dry pavement, find the stopping distance from 28.7 m/s with locked brakes. (b) With $\mu_k = 0.25$ on wet pavement, find the initial speed that gives the same distance.

(a) In this sliding-friction model on a level road, $N = mg$ and the acceleration is $a = -\mu_k g$. The stopping distance follows from $0 = v_o^2 + 2ad$.

$$d = v_o^2 / (2\mu_k g) = (28.7)^2 / [2(0.80)(9.80)]$$

$$d = \mathbf{52.5 \text{ m}} \text{ (about 53 m)}$$

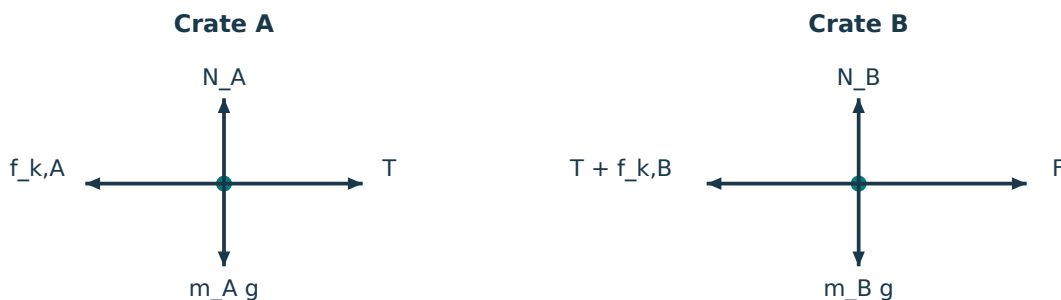
(b) Set the wet and dry stopping distances equal:

$$v_{\text{wet}}^2 / \mu_{\text{wet}} = v_{\text{dry}}^2 / \mu_{\text{dry}}$$
$$v_{\text{wet}} = 28.7 \sqrt{(0.25/0.80)} = \mathbf{16.0 \text{ m/s}} \text{ (about 36 mi/h)}$$

These are braking distances only; reaction distance is excluded. The textbook notes that locking the brakes is not the safest stopping method.

5.37 Connected crates on a horizontal floor

Crates A and B have masses m_A and m_B , connected by a rope. Both have kinetic-friction coefficient μ_k . A horizontal force F pulls B to the right at constant velocity. Draw force diagrams and find (a) F and (b) the connecting-rope tension.



On B, the leftward arrow represents two separate forces, T and $f_{k,B}$. Vertically, $N_A = m_A g$ and $N_B = m_B g$.

(a) Treat both crates as one system. The rope tension is internal and cancels. With $a = 0$:

$$F - \mu_k m_A g - \mu_k m_B g = 0$$

Answer (a): $F = \mu_k(m_A + m_B)g$.

(b) Isolate crate A. Its rightward tension balances its leftward friction:

$$T - \mu_k m_A g = 0$$

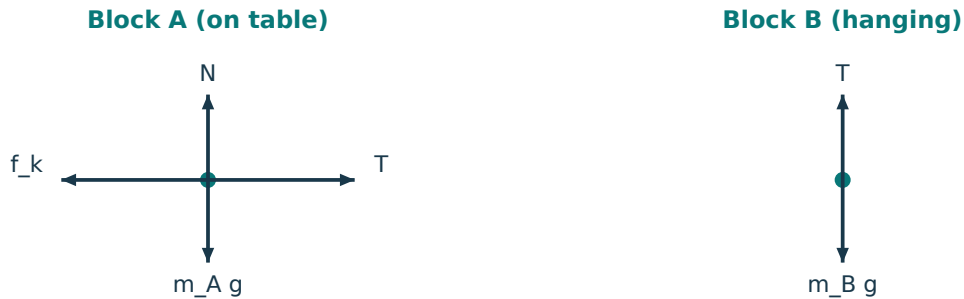
Answer (b): $T = \mu_k m_A g$.

Friction • Worked Solutions

5.39 • Table-and-hanging-block system | 5.41 • Pushing downward at an angle

5.39 Two blocks connected over a pulley

Block A (2.25 kg) sits on a table with $\mu_k = 0.450$. A light cord over a frictionless pulley connects it to hanging block B (1.30 kg). Released from rest, find (a) each speed after moving 3.00 cm and (b) the cord tension.



Assume the blocks begin sliding as the problem intends. Take rightward for A and downward for B as positive. The cord gives both blocks the same acceleration magnitude.

$$A: T - \mu_k m_A g = m_A a$$

$$B: m_B g - T = m_B a$$

$$a = g(m_B - \mu_k m_A) / (m_A + m_B) = 0.7937 \text{ m/s}^2$$

(a) $v = \sqrt{(2a\Delta x)} = \sqrt{[2(0.7937)(0.0300)]} = \mathbf{0.218 \text{ m/s}}$ for each block. A moves toward the pulley; B moves downward.

(b) $T = m_B(g - a) = 1.30(9.80 - 0.7937) = \mathbf{11.7 \text{ N}}$.

5.41 A downward push increases friction

A crate of mass m has friction coefficients μ_s and μ_k on a level floor. A woman pushes at angle θ below horizontal. (a) Find F for constant velocity. (b) Find the critical μ_s at which no finite push can start it moving.

Forces: weight mg downward; normal N upward; push components $F \cos \theta$ forward and $F \sin \theta$ downward; friction backward.

(a) Vertical balance gives $N = mg + F \sin \theta$. For steady forward sliding:

$$F \cos \theta = \mu_k N = \mu_k (mg + F \sin \theta)$$

Answer (a): $F = \mu_k mg / (\cos \theta - \mu_k \sin \theta)$.

A positive finite force requires $\cos \theta - \mu_k \sin \theta > 0$; otherwise this push cannot sustain constant forward sliding.

(b) Starting motion requires the horizontal push to exceed maximum static friction:

$$F \cos \theta > \mu_s (mg + F \sin \theta)$$

$$F(\cos \theta - \mu_s \sin \theta) > \mu_s mg$$

If $\mu_s \geq \cot \theta$, the left coefficient is nonpositive, so no finite positive F can satisfy the inequality ($0^\circ < \theta < 90^\circ$).

Answer (b): $\mu_{s,\text{critical}} = \cot \theta = 1/\tan \theta$.